

Remaining schematic work

PetSafe 100-1339 | v0.9.31 | 2026-10-09

12 active issues. Completed steps are archived, not bench requests. Next: six directional Q3 diode readings using docs/FINISHING_MEASUREMENTS.html.

Drawing / connectivity	Remaining data	ERC
0 fitted open pads; 0 open PIC pads; 0 dangling wire ends. 83 native/model net partitions match.	44 ceramic values; LED1 footprint blank. Three local-only PIC branches and R33 inventory remain explicit.	1 retained U6/U6A output conflict. Five stock power-source flags added; physical net partitions unchanged.

Audit boundary

Native wire geometry, library definitions, all catalog pad assignments and existing evidence were checked. Zero open pads does not establish hidden copper or functional behavior. U6A is DNP; its regulator symbol and visible conflict remain by user choice. No new NC markers, net merges or ERC exclusions.

Order of work

Resolve hidden paths and device behavior: One Q3 diode batch next; R33 inventory and three local-only PIC branches remain explicit. Other checks are staged, not requested together. Use the next measurement screen. Agent resolves visible copper and prepares only discriminating hidden-path or operating tests.

Recover critical component values: 44 ceramic values; prioritize tuning and receiver after topology. Use the LCR meter with known test conditions and account for parallel components.

Finish interface and package mapping: LED channel/pad map and footprint; motor wire order. Exact physical fit is a later PCB qualification. Use LED diode tests and pad geometry; retain measured body sizes.

Resolve hidden paths and device behavior

E03 - RF supply and Q8 drive

Established: C5/C6 parallel wiring and Q8 ground/output joins are established. SIL3724A and the U2 inverter remain candidate identities.

Remaining: The Q8 source supply path has a 10-ohm result rather than proven copper continuity. P-channel gate turn-off margin and RF timing remain unmeasured.

Next: First confirm Q8 physical pin2 to C6 supply end with stable probe contact if the photo path remains ambiguous. In the later powered session compare source and gate levels; a multimeter cannot establish RF gate timing.

Close when: Source path and actual gate-to-source drive are supported; C6 value is handled by V01.

E04 - Antenna and tuning-cell topology

Established: Five PIC-to-base-resistor routes and the visible Q3/R13/C10 joins are mapped. Seven tested midpoint pairs are isolated (OL).

Remaining: PNP emitter-at-GND is not ordinary GPIO-controlled PNP switching. Device polarity/base mapping or the assumed RF mechanism may be wrong. The five capacitor midpoints lack an established antenna continuation.

Next: Next batch: six directional diode-mode readings on Q3 only, using the labelled photo. This distinguishes PNP/NPN/base candidates; collector/emitter and hidden antenna links may still need one follow-up. No repeated visible-trace tests.

Close when: One cell has a supported device map and functional antenna path; the other four are checked only where needed.

E05 - Button contact behavior

Established: R41 is 330 ohm to J1 VPP. The switch-side node also reaches R40 and C35; opposite C35 side is GND.

Remaining: Internal contact pairings and released/pressed behavior of this switch have not been measured.

Next: Later unpowered check: measure across the two right-hand S1 legs, released then pressed, and each same-row left/right pair released. This checks the four-pad mapping without repeating any resistor routing.

Close when: Same-row pairs and open/closed switching agree with the drawing, or the switch pin mapping is corrected.

E06 - PIC branches without an established onward endpoint

Established: PIC2/RA0 reaches C25/V031; PIC14/RC3 reaches TP11; PIC18/RC7 reaches TP17. RF common-drive wiring is now photo-supported.

Remaining: RA0/C25, RC3/TP11 and RC7/TP17 may terminate locally or continue internally. R3/R4/C28-to-RB3 and D3/R42-to-RB4 circuit roles are unverified.

Next: Review matching front/rear via locations before proposing candidate continuity pairs. TP16-to-TP11/TP17 was already rejected; do not repeat it. No generic all-to-all test is requested.

Close when: Each of the three local-only branches has a traced continuation or explicit evidence of intentional termination; functional roles are qualified separately.

E07 - PIR: R33 inventory and Q2 behavior

Established: J3/U7 ordering, U6 external pads and PIC21/TP16/Q2 output routing are established. U6 pin5 is tied to VIN; pin4 is unused.

Remaining: R33 is inherited as a fitted 10k PIR pull-up, but its cited IMG_2435 shows R37/R38/R39 and does not establish R33. Its existence/location and both assumed endpoints must be resolved. Q2 pin functions and actual PIR voltage also remain unverified.

Next: Locate R33 before requesting measurements of it. If a real R33 exists, identify its pads and check its two proposed nets (J3.2 and J3.1). If the inventory duplicated another resistor, remove that unsupported entry after corroboration. Q2 behavior belongs in the later powered check.

Close when: R33 is physically accounted for or withdrawn with evidence; Q2 signal behavior and U6 output are supported. No repeat connector or NC-pad checks.

E08 - Receiver and Q7 operating behavior

Established: U3 input returns and R23/R42/C40, C23/R25/R48 and DNP R47/C42 paths are reconciled. VREF supplies U3 pin5.

Remaining: Q7 transistor pin roles/supply switching, receiver gain/filter response and the functional MCU interface are unverified. H_RX_DETECT is only a series junction, not proof of a missing output wire.

Next: Do not invent another detector wire. In one powered session measure VDD, U3 pin8, VREF and U3 outputs pins1/7, idle and active. Interpret waveforms only after critical capacitor values are recovered.

Close when: Receiver power and bias are supported and the existing PIC25/RA1 interfaces have a documented role; no presumed output destination is used as a closure criterion.

E09 - Supply voltages and Q1 control

Established: BATTERY+, post-Q1 VSYS, VDD and VREF are separate nets. L1/L2 supply sides share VSYS. Five ERC source declarations now describe existing supply paths.

Remaining: Actual voltages and Q1 gate behavior have not been measured. Candidate regulator ratings are not measured board voltages.

Next: Later powered session using the original battery: BATTERY+, VSYS, VDD, VREF, U6 output, Q1 gate and switched receiver supply, idle and active. Keep Q8 timing with E03.

Close when: Actual supply levels and control behavior are recorded without merging battery, VSYS or filtered branches.

E10 - D6 breakdown and ICSP compatibility

Established: D6 anode is GND and cathode is VPP; forward drop is about 0.7V and reverse diode mode is OL. J1 pin order is resolved.

Remaining: G3 does not uniquely establish breakdown voltage. The displayed 3.9V candidate is not a measured clamp rating and could conflict with programming voltage.

Next: No more continuity or ordinary diode tests. Agent seeks a matching primary marking specification; if still ambiguous, plan a current-limited isolated breakdown test. Do not apply a high-voltage test through the PIC.

Close when: Clamp class/rating is established or explicitly bounded for reconstruction, and programming compatibility is checked.

I01 - Candidate device identities and pin roles

Established: Manufacturer candidate datasheets and pin tables exist. D2 local wiring is complete; measured R -> S -> L diode behavior supports a series pair.

Remaining: Exact fitted identity/ratings remain candidates for D1-D6, Q1-Q8/Q11, U1/U2/U6. The 4P marking alone does not identify three-pin D2; a manufacturer 4P code for a two-pin diode is incompatible.

Next: Resolve electrical risks through E03/E04/E07/E08/E09/E10. For remaining ambiguity retain the supported equivalent circuit and disclose ratings not established; no sharper-marking request or repeat D2 rail sweep. Exact manufacturer recovery is not always possible with a meter.

Close when: Each part has a supported pin/function model and adequate ratings or an explicit accepted-equivalent limitation; exact branding is not falsely claimed.

Recover critical component values

V01 - Capacitor values and passive ratings

Established: C5 is 470uF/16V. L1/L2 readings are 1.4/2.2uH. Marked resistor nominal values are recorded; R33 inventory is separately under E07.

Remaining: 44 fitted ceramic values remain unknown. L1/L2 measurement conditions and component ratings are not established.

Next: After tuning topology, prioritize antenna/RF/receiver capacitors with the LCR meter. Record frequency, mode and isolation. Parallel parts give a combined reading; lift one terminal only where an individual value is needed. Report existing L1/L2 measurement conditions without remeasuring yet.

Close when: Critical individual values are recovered; any bypass substitutions and unverified ratings are explicitly identified assumptions.

All unrecovered ceramic values

C1, C2, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C43, C44, C45, C46, C47, C48

Finish interface and package mapping

I02 - LED channel map and motor lead orientation

Established: LED1 is red/green in a 1.5 mm square body; two control branches and common ground are mapped. U5 motor outputs are modeled.

Remaining: Which resistor branch lights red/green and exact LED pad numbering remain unknown. Physical motor lead A/B correspondence is provisional.

Next: Later diode-mode check: red probe on each resistor-fed LED contact and black on its common-ground contact; record colour/voltage or OL. Match motor lead positions to U5 output5/output8 only if not visibly traceable.

Close when: LED internal channels/polarity and motor connector physical ordering are supported.

M01 - LED footprint and mechanical qualification

Established: 148/149 candidate footprints assigned. C5 diameter/height, S1 body/actuator size, R8 and C11 sizes are already measured.

Remaining: LED1 land pattern/numbering is the only blank footprint. C5 pitch and S1 pad fit/3D heights are reproduction qualifications, not unknown schematic wires.

Next: Obtain LED underside/pad geometry or a matching vendor drawing before selecting a footprint. For later PCB reproduction check C5 lead pitch and S1 pad fit; use actual 16mm capacitor and 6mm switch heights. Do not repeat body dimensions.

Close when: LED footprint has a supported pad map; remaining package-fit assumptions are documented separately from schematic connectivity.

Scope and sources

Main-board schematic finalization. Routed PCB/layer stack, PIR daughterboard internals and firmware reconstruction remain later work, not completed. Empty-option population design is not needed to reproduce the observed DNP board.

DNP options retain all modeled local routes and candidate roles. J6.2/R36.1 terminates at an unpopulated external interface; no additional onboard rail is assumed. U6A stays a regulator symbol with its visible output conflict by explicit user choice.

All previous issue dispositions: evidence/finalization_review_v0931.json. Native geometry and uncertainty coverage: evidence/finalization_audit.json. Current register: evidence/remaining_work.json. Archived observations and old checklists: docs/history/README.md.

Power-source flags describe supply paths through existing passive components; they do not prove voltage or transistor operation. Installed KiCad 10 libraries and project symbols match the embedded pin/graphic definitions. GUI library reload has not been verified.

Reference rails

Rail	Existing modeled path
BATTERY+	Battery positive -> Q1 single pad/native3 (candidate drain); TP14/R43.2 on the raw side.
VSYS	Q1.L/native2 -> U5/MX512H pin4; L1.1 and L2.1 supply sides.
VDD	TP103, PIC20/C32, U5 pin1/VCC, V064/V063/V075/V082. D3.R via V082 is user-confirmed.
VREF	TP104, PIC4/RA2/V035/C39, R21.2/V071. U3.5 and R22 D/left/model2 at 1 ohm.
GND	Battery negative and ground copper.